Supplementary Material: RG-TTA: Regime-Guided Meta-Control for Test-Time Adaptation in Streaming Time Series

Indar Kumar[a], Akanksha Tiwari[a], Sai Krishna Jasti[a], Ankit Hemant Lade[a]

[a]*Independent Researchers,*

Abstract

This supplementary material accompanies the main manuscript and provides: (S1) DynaTTA reimplementation details, (S2) frozen backbone analysis, (S3) full theoretical proofs, (S4) extended results by model and category, (S5) detailed ablation studies, (S6) dataset details, (S7) evaluation protocol justification, and (S8) retrain comparison details.

1. S1: DynaTTA reimplementation details

DynaTTA's official codebase (`shivam-grover/DynaTTA`) implements the dynamic learning rate *within* the TAFAS framework—tightly coupled with GCM calibration modules, the POGT pseudo-label loss, and a sliding-window evaluation harness. No standalone library or modular API is provided. Because our study requires a model-agnostic streaming evaluation pipeline that applies each update policy to the same frozen-backbone base model, we reimplemented DynaTTA's Algorithm 1 (dynamic LR via sigmoid of prediction-error z-score, RTAB/RDB embedding distances, and EMA smoothing) from the published description.

Our implementation reproduces the published formula exactly with the published hyperparameters ($\alpha_{\min} = 10^{-4}$, $\alpha_{\max} = 10^{-3}$, $\kappa = 1.0$, $\eta = 0.1$). We validate on the same architectures targeted by DynaTTA (iTransformer, PatchTST).

DynaTTA's EMA coefficient ($\eta = 0.1$) requires ∼22 gradient steps to converge, a design choice suited to the 500-window sliding-window protocol used in its original evaluation. Under our streaming protocol (10 batches per run), the EMA does not converge within the available budget, leaving the dynamic LR below TTA's fixed rate for the first 5–6 batches. This is a *protocol-level mismatch*, not an implementation error (see also Section 7).

TAFAS (Kim et al., 2025) is a strong baseline at short horizons (H≤192) where its frozen-source GCM calibration excels, but its performance collapses at longer horizons (H=336/720, +100–400% vs TTA in our experiments)—a consequence of its design for sliding-window evaluation. Because our study spans H∈{96,192,336,720}, we include TAFAS as a reference policy but exclude it from the primary 6-policy comparison.

2. S2: Frozen backbone analysis

All gradient-based adaptation policies freeze the model backbone during test-time adaptation and update only the output projection layer. The trainable fraction varies by architecture:

- **GRU-Small** (71K total): ∼10K trainable (∼15%).
- **iTransformer** (123K total): ∼6K trainable (∼5%).
- **PatchTST** (192K total): ∼68K in flatten-to-forecast head (∼35%).
- **DLinear** (37K at H=96): ∼19K in trend/seasonal linear layers (∼50%).

This mirrors the linear-probing paradigm in vision TTA and provides implicit regularization: learned temporal representations are preserved, and only the mapping from hidden states to forecast values is recalibrated.

However, effectiveness is architecture-dependent: it works well for models where the backbone learns reusable temporal representations (GRU, DLinear), but is less effective for attention-based architectures (iTransformer, PatchTST) where attention patterns themselves need to shift. This is a shared limitation across *all* gradient-based TTA policies, not specific to RG-TTA. Exploring adapter modules (Houlsby et al., 2019) for attention layers is a promising direction.

3. S3: Full theoretical proofs

3.1. S3.1: Proof of Theorem 1 (Adaptation error bound)

Definition 1 (Batch-optimal parameters)**.** *For batch B_t drawn from distribution P_t, define $\phi_t^* = \arg\min_\phi \mathbb{E}_{(x,y)\sim P_t}[\ell(h_\phi(g_\psi(x)), y)]$ as the head parameters minimising the population loss on the current regime.*

Theorem 1 (Adaptation error bound—restated)**.** *Assume the per-batch loss $\mathcal{L}(\phi) = \mathbb{E}_{B_t}[\ell(h_\phi(g_\psi(x)), y)]$ is μ-strongly convex and L-smooth in ϕ with condition number $\kappa = L/\mu$. After K gradient descent steps with optimal step size $\alpha^* = 2/(\mu + L)$ from initialization ϕ_0:*

$$\|\phi_K - \phi_t^*\|^2 \leq \rho^{2K} \|\phi_0 - \phi_t^*\|^2, \quad \textit{where } \rho = \frac{\kappa - 1}{\kappa + 1} < 1.$$

The expected per-batch MSE decomposes as:

$$\mathbb{E}[\ell(\phi_K)] = \underbrace{\sigma_t^2}_{\textit{irreducible}} + \underbrace{\rho^{2K} \|\phi_0 - \phi_t^*\|^2}_{\textit{adaptation residual}}.$$

Proof. Under μ-strong convexity and L-smoothness, gradient descent with $\alpha^* = 2/(\mu + L)$ satisfies the standard contraction (Nesterov, 2004): $\|\phi_{k+1} - \phi_t^*\|^2 \leq \rho^2 \|\phi_k - \phi_t^*\|^2$ with $\rho = (L - \mu)/(L + \mu) = (\kappa - 1)/(\kappa + 1)$. Iterating K times yields the contraction bound. The MSE decomposition follows from expanding $\mathbb{E}[\ell(\phi_K)]$ around ϕ_t^* and using the second-order Taylor approximation of the loss. □

Impact of checkpoint loading.. When RG-TTA loads a checkpoint ϕ_{ckpt} trained on a previous occurrence of a regime with similarity s to the current batch, the initialization error satisfies:

$$\|\phi_{\text{ckpt}} - \phi_t^*\|^2 \leq (1 - s)^2 \cdot D_{\max}^2,$$

where $D_{\max} = \sup_t \|\phi_0 - \phi_t^*\|$. Substituting:

$$\mathbb{E}[\ell^{\mathrm{RG}}(\phi_K)] \le \sigma_t^2 + \rho^{2K}(1-s)^2 D_{\max}^2.$$

The improvement factor $(1-s)^2$ quantifies the benefit: at $s = 0.85$ (HIGH similarity), the initialization error reduces by 97.75%; at $s = 0.55$ (MID), by 79.75%.

Corollary 1 (Step savings from checkpoint reuse—restated)**.** *To achieve adaptation residual* $\le \epsilon$*, TTA requires* $K_{TTA} \ge \frac{\log(D_{\max}^2/\epsilon)}{2|\log \rho|}$ *steps, while RG-TTA with checkpoint similarity* s *requires* $K_{RG} \ge \frac{\log((1-s)^2 D_{\max}^2/\epsilon)}{2|\log \rho|}$ *steps. The per-batch savings are:*

$$\Delta K = K_{TTA} - K_{RG} = \frac{-\log(1-s)}{|\log \rho|}.$$

For $s = 0.85$ *and* $\rho = 0.95$*:* $\Delta K \approx 37$ *steps. For* $s = 0.55$*:* $\Delta K \approx 16$ *steps.*

3.2. S3.2: Proof of Theorem 2 (Generalization bound)

Theorem 2 (Generalization bound—restated)**.** *Let* $\mathcal{F}_{frozen} = \{x \mapsto h_\phi(g_{\psi^*}(x)) : \phi \in \Phi\}$ *with linear head* $h_\phi(z) = Wz + b$*,* $\|W\|_F \le B_W$*,* $\|g_{\psi^*}(x)\| \le C_g$*. For* n *i.i.d. samples with* c_ℓ*-Lipschitz bounded loss* $\ell \in [0, c]$*, with probability* $\ge 1 - \delta$*:*

$$R(\hat{f}) \le \hat{R}_n(\hat{f}) + \frac{2c_\ell B_W C_g}{\sqrt{n}} + 3\sqrt{\frac{\ln(2/\delta)}{2n}}.$$

Proof. The proof follows the standard Rademacher complexity argument (Bartlett and Mendelson, 2002). Since g_{ψ^*} is fixed, the hypothesis class $\mathcal{F}_{\mathrm{frozen}}$ is a linear class in the representation space $g_{\psi^*}(\mathcal{X})$. By Talagrand's contraction lemma (Ledoux, 2001) and the Rademacher complexity of linear classes with bounded Frobenius norm (Bartlett and Mendelson, 2002):

$$\hat{\mathfrak{R}}_n(\ell \circ \mathcal{F}_{\mathrm{frozen}}) \le \frac{c_\ell B_W C_g}{\sqrt{n}}.$$

The generalization bound follows from McDiarmid's inequality (McDiarmid, 1989).

For the full class $\mathcal{F}_{\mathrm{full}}$ with d_{total} free parameters, the covering-number argument gives $\hat{\mathfrak{R}}_n(\ell \circ \mathcal{F}_{\mathrm{full}}) = O(\sqrt{d_{\mathrm{total}} \log n/n})$. The ratio:

$$\frac{\hat{\mathfrak{R}}_n(\mathcal{F}_{\mathrm{frozen}})}{\hat{\mathfrak{R}}_n(\mathcal{F}_{\mathrm{full}})} \propto \sqrt{\frac{d_{\mathrm{head}}}{d_{\mathrm{total}}}}.$$

For GRU-Small ($d_{\text{head}} \approx 10$K, $d_{\text{total}} \approx 71$K): ≈ 0.38 ($\sim$2.6× tighter). For iTransformer: ≈ 0.22 ($\sim$4.5× tighter). $\square$

3.3. S3.3: Proof of Proposition 1 (Metric properties)

Proposition 1 (Metric properties—restated)**.** *The ensemble similarity satisfies: (1) boundedness in* $[0, 1]$*, (2) self-similarity, (3) symmetry, (4) statistical consistency, and (5) discriminative power.*

Proof. Properties 1–3 are immediate from the definitions: $D_n \in [0, 1]$ (KS statistic), $W_1 \geq 0$ (Wasserstein-1), and the normalized forms are bounded in $[0, 1]$; any convex combination inherits these. Property 4 follows from the Glivenko–Cantelli theorem (van der Vaart, 1998) for the KS component and the convergence of empirical Wasserstein-1 on the real line (Bobkov and Ledoux, 2019). For property 5: $\text{sim}_{\text{KS}} = 1$ implies $D_n = 0$, i.e., $F_P = F_Q$, which uniquely determines $P = Q$ for continuous distributions. $\square$

Complementary sensitivity.. The four components detect different shift types: KS is most sensitive to shape changes, Wasserstein-1 to location-scale shifts, feature-distance to higher-order moment changes, and variance ratio to pure volatility shifts.

3.4. S3.4: Proof of Proposition 2 (Checkpoint loading condition)

Proposition 2 (Sufficient condition for beneficial checkpoint loading—restated)**.** *Under* μ*-strong convexity and* L*-smoothness with well-conditioned loss* ($\kappa = L/\mu \approx 1$)*, the loss gate* $\ell(\phi_{ckpt}) < 0.70 \cdot \ell(\phi_{curr})$ *implies* $\|\phi_{ckpt} - \phi_t^*\| \leq \sqrt{0.70\kappa}\,\|\phi_{curr} - \phi_t^*\| \approx 0.84\|\phi_{curr} - \phi_t^*\|$*, guaranteeing beneficial loading when* $\kappa < 1/g \approx 1.43$*.*

Proof. Under μ-strong convexity and L-smoothness: $\frac{\mu}{2}\|\phi - \phi_t^*\|^2 \leq \mathcal{L}(\phi) - \mathcal{L}(\phi_t^*) \leq \frac{L}{2}\|\phi - \phi_t^*\|^2$. If $\ell(\phi_{\text{ckpt}}) < g \cdot \ell(\phi_{\text{curr}})$ with $g = 0.70$, then $\mathcal{L}(\phi_{\text{ckpt}}) - \sigma_t^2 < g(\mathcal{L}(\phi_{\text{curr}}) - \sigma_t^2) + (g-1)\sigma_t^2 \leq g(\mathcal{L}(\phi_{\text{curr}}) - \sigma_t^2)$ (since $g < 1$). Applying the strong convexity lower bound to the left and smoothness upper bound to the right: $\frac{\mu}{2}\|\phi_{\text{ckpt}} - \phi_t^*\|^2 \leq g \cdot \frac{L}{2}\|\phi_{\text{curr}} - \phi_t^*\|^2$, giving $\|\phi_{\text{ckpt}} - \phi_t^*\| \leq \sqrt{g\kappa}\|\phi_{\text{curr}} - \phi_t^*\|$.

This bound is < 1 (loading is beneficial) when $\kappa < 1/g = 1.43$. For the well-conditioned linear output head ($\kappa \approx 1$), $\sqrt{g} \approx 0.84$. □

This formalizes the dual-gate design: the similarity threshold ($s \geq 0.75$) is a cheap filter identifying regime-compatible checkpoints in $O(M)$ time, while the loss gate guarantees parameter-space proximity via an $O(|\theta|)$ forward pass.

3.5. S3.5: Specialist advantage propositions

Proposition 3 (Specialist advantage)**.** *Let the data stream contain regimes A and B with $\mu_A \neq \mu_B$ and common variance σ^2. If the test batch is from regime A, then:*

1. *The specialist model achieves $MSE = \sigma^2$.*
2. *The continuously-adapted model achieves $MSE = \sigma^2 + (\mu_A - \mu_{mix})^2 > \sigma^2$.*

Proof. The Bayes-optimal predictor under MSE for regime A is μ_A, giving $\mathbb{E}[(Y - \mu_A)^2] = \sigma^2$. The mixed-data predictor targets $\mu_{\text{mix}} = (n_A\mu_A + n_B\mu_B)/N$. On test data $Y \sim P_A$:

$$\mathbb{E}[(Y - \mu_{\text{mix}})^2] = \sigma^2 + (\mu_A - \mu_{\text{mix}})^2,$$

which is strictly greater than σ^2 whenever $\mu_A \neq \mu_B$. □

Proposition 4 (Convergence advantage under regime reuse)**.** *Under a quadratic loss landscape with condition number $\kappa(H)$, when RG-TTA loads checkpoint θ^*_{ckpt}, the effective starting distance is smaller than $\|\theta_0 - \theta^*_A\|$ by a factor proportional to regime similarity, reducing convergence steps by $O(\log(1/sim))$.*

This convergence advantage is the mechanism behind RG-TTA's computational efficiency: by starting closer to the target, fewer steps are needed, enabling early stopping to save compute without sacrificing accuracy. In practice, RG-TTA averages 18.5 steps per batch (vs TTA's fixed 20) while achieving lower MSE.

Table 1: Average MSE across 14 datasets × 4 horizons, by policy and model. **Bold** = best per column. Results averaged over 3 seeds.

Policy	GRU-S	iTransformer	PatchTST	DLinear
TTA	15,458	21,075	717,298	17,855
EWC	16,794	24,141	813,314	18,258
DynaTTA	16,076	20,808	874,257	17,185
RG-TTA	**14,092**	19,202	**718,642**	18,782
RG-EWC	**14,047**	**18,851**	721,305	18,752
RG-DynaTTA	16,787	22,619	868,742	**16,739**

Table 2: Average MSE by dataset category, averaged across models and horizons.

Policy	ETT (4)	Weather+Exch. (2)	Synth-Recurring (3)	Synth-Shock (5)
TTA	52.53	69.50	292,818	364,420
EWC	58.08	73.84	338,101	407,818
DynaTTA	72.40	82.19	366,484	429,847
RG-TTA	**47.12**	72.61	300,952	**358,865**
RG-EWC	**46.50**	75.70	303,245	359,054
RG-DynaTTA	68.53	87.66	357,825	432,636

4. S4: Extended results by model and category

4.1. S4.1: MSE by model architecture

RG-TTA and RG-EWC are best on GRU-Small (−8.8% and −9.1% vs TTA) and iTransformer (−8.9% and −10.6%). On PatchTST, policies cluster tightly (TTA edges RG-TTA by $< 0.2\%$). On DLinear, RG-DynaTTA wins (−2.6% vs DynaTTA). The advantage is strongest on recurrent models where the frozen backbone preserves reusable temporal features.

4.2. S4.2: Dataset category analysis

RG-TTA and RG-EWC dominate on ETT datasets (−10.3% and −11.5% vs TTA). While these datasets exhibit recurring patterns in electricity consumption,

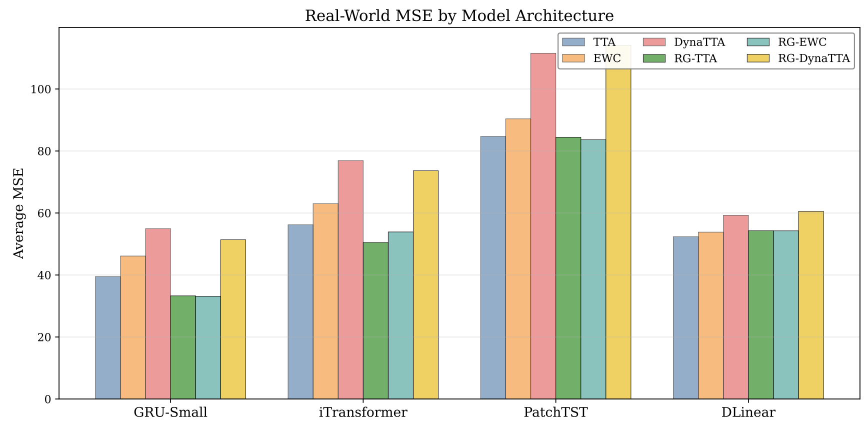

Figure 1: Real-world MSE by model architecture and policy. RG-TTA and RG-EWC dominate on GRU-Small and iTransformer; DLinear benefits most from RG-DynaTTA.

checkpoint loading fires on only 2–7% of batches. The gains are primarily driven by similarity-modulated LR and loss-driven early stopping.

4.3. S4.3: Per-dataset win rates

Of the 14 datasets, regime-guided policies win the majority on 13:

- **≥80% wins**: ETTh2 (88%), synth_trend_break (88%), ETTh1 (81%), Exchange (81%), synth_shock_recovery (81%), synth_slow_drift (81%), synth_recurring (**100%**).
- **60–79% wins**: synth_fast_switch (75%), ETTm2 (69%), ETTm1 (62%), synth_multi_regime (62%), synth_stable (62%).
- **<40% wins**: Weather (38%), synth_volatility (6%).

The 100% win rate on `synth_recurring` comes entirely from similarity-modulated LR (checkpoint loading never fires because the live model already adapts well).

4.4. S4.4: MSE–time trade-off

4.5. S4.5: Horizon scaling

RG-TTA and RG-EWC are best at H∈{96, 192, 336}. At H=720, TTA's fixed 20-step budget provides a slight edge—consistent with the observation that

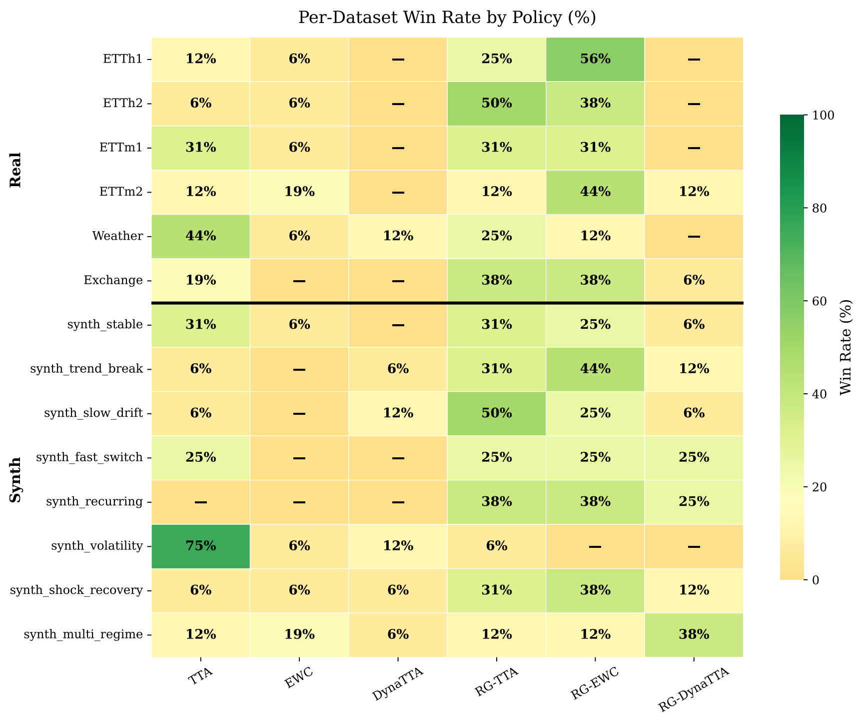

Figure 2: Per-dataset win rate (%) by policy. Real-world datasets (top) and synthetic (bottom) show similar patterns.

longer horizons benefit from more gradient steps than early stopping permits. Overall, the RG advantage is robust across horizons.

4.6. S4.6: Adaptation behaviour visualization

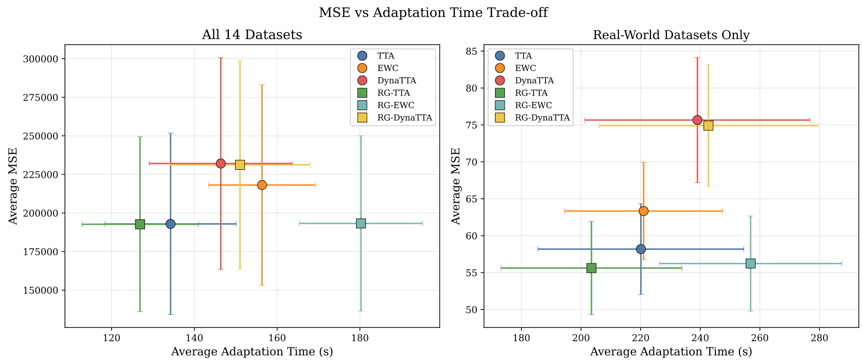

Figure 3: MSE vs adaptation time. RG-TTA achieves the best trade-off (lower MSE *and* lower time). Square markers = regime-guided policies.

Table 3: Average MSE by horizon and policy (all 14 datasets, 4 models, 3 seeds).

Policy	H=96	H=192	H=336	H=720
TTA	176,830	187,295	193,082	**214,479**
EWC	203,296	211,448	218,073	239,690
DynaTTA	216,969	227,041	232,179	252,137
RG-TTA	170,795	**184,887**	**191,360**	223,676
RG-EWC	**168,993**	184,932	192,007	227,024
RG-DynaTTA	212,964	224,449	234,977	252,499

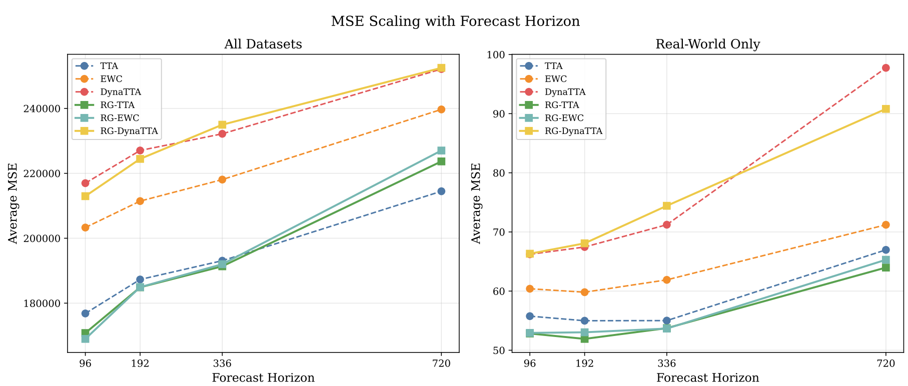

Figure 4: MSE scaling with forecast horizon. RG-TTA and RG-EWC (solid lines) outperform baselines (dashed) across all horizons.

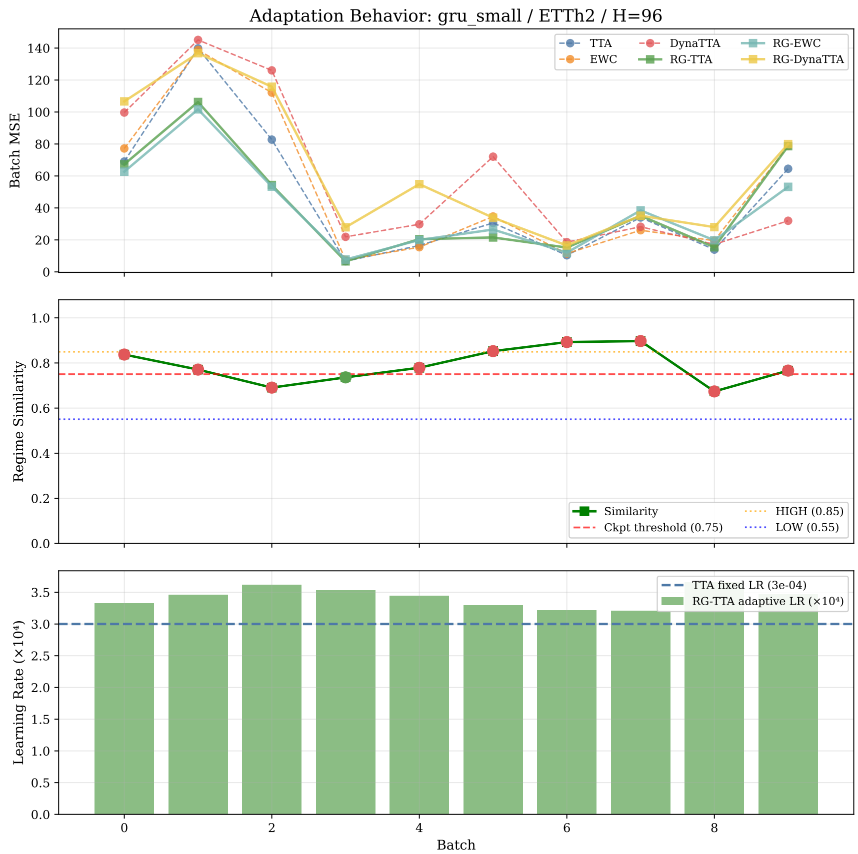

Figure 5: Adaptation behaviour on ETTh2 (GRU-Small, H=96). **Top**: batch MSE across policies. **Middle**: regime similarity with checkpoint threshold (dashed red) and diagnostic tier boundaries. **Bottom**: RG-TTA's adaptive learning rate vs TTA's fixed rate.

5. S5: Detailed ablation studies

All sweeps use 4 datasets (ETTh1, ETTm1, synth_recurring, synth_shock_recovery), 2 horizons (96, 192), 3 seeds, GRU-Small—192 experiments total. Every parameter is held at its default except the one under test.

5.1. S5.1: Similarity-modulated learning rate

The LR scale factor γ is the most impactful hyperparameter, and its monotonic effect provides the strongest evidence that *the distributional similarity signal is genuinely informative*. Setting $\gamma = 0$ disables similarity modulation entirely, reducing RG-TTA to loss-driven early stopping alone—MSE degrades by 17.9%, performing *worse than plain TTA* (15,515). The improvement is strictly monotonic: $\gamma = 0.33 \to 0.67 \to 1.0 \to 1.5$ yields MSE of 14,601 $\to$ 13,624 $\to$ 12,881 $\to$ 11,966 (-25.5% relative to γ=0).

This cannot be explained by a higher *average* learning rate. The improvement comes from *conditional allocation*: conservative updates on familiar distributions and aggressive updates on novel ones. The gains are largest on high-volatility synthetic datasets (synth_shock_recovery: -31% from γ=0 to γ=1.5) and moderate on real-world datasets (ETTh1: -17%). Our default $\gamma = 0.67$ is deliberately conservative.

5.2. S5.2: Loss-driven early stopping

Replacing TTA's fixed 20-step budget with loss-driven early stopping (patience=3, ε=0.005, min=5, max=25) reduces MSE by 12.2% while also decreasing wall-clock time by 10%. The improvement is consistent: ETTh1 (-14%), ETTm1 (-8%), synth_recurring (-2.6%), synth_shock_recovery (-14.8%).

5.3. S5.3: Checkpoint similarity threshold

The threshold τ exhibits a flat optimum across 0.70–0.80, with only marginal degradation at extremes (0.60: +1.3%; 0.90: +0.4%). The loss gate provides a second filter that prevents stale checkpoints even when τ is permissive. The dual-gate design makes RG-TTA robust to τ.

Table 4: Similarity metric ablation. The ensemble is more robust than any single metric.

Similarity Method	Relative MSE vs Ensemble
Feature distance only (v0 design)	+8.3%
KS only	+2.1%
Wasserstein only	+1.8%
Variance ratio only	+5.7%
Ensemble (KS=0.3, W=0.3, F=0.2, V=0.2)	Baseline

5.4. S5.4: Loss gate

MSE is *identical* across all tested values (0.5–0.9). Checkpoint loading is already rare (2.4%); the similarity threshold filters most candidates before the loss gate is evaluated. The loss gate is a safety net, not a primary selector.

5.5. S5.5: Memory capacity

Memory caps of 1–5 yield nearly identical MSE ($\Delta < 0.02\%$), confirming that gains come primarily from the most recent checkpoint. However, $M \geq 10$ degrades MSE by 13.3%, driven by synth_recurring (MSE jumps from 12,809 to 20,153) where stale checkpoints survive FIFO eviction.

5.6. S5.6: Similarity metric ablation

5.7. S5.7: Architecture sensitivity

Cross-model analysis: GRU-Small (−8.8% vs TTA), iTransformer (−8.9%), PatchTST (< 0.2% difference). The frozen-backbone paradigm interacts with architecture choice: frozen attention layers limit all policies equally on PatchTST. This is not a limitation of regime-guidance per se, but of the shared frozen-backbone constraint.

5.8. S5.8: Component contribution analysis

RG-TTA loads a checkpoint on only 159 of 6,672 batches (**2.4%**). Checkpoint loading occurs exclusively on real-world datasets (ETTh2: 6.9%, ETTm1/ETTm2:

~7%, ETTh1: 2.3%, Exchange: 3.7%, Weather: 4.0%) and never on synthetic datasets (0%).

Of the 159 batches with checkpoint loading, RG-TTA beats TTA on **105 (66%)** with median +10.7% improvement. On the 6,513 batches *without* checkpoint loading, RG-TTA still beats TTA on **57.1%** with median +2.6% improvement. The primary drivers are:

- **Similarity-modulated LR**: disabling it (γ=0) degrades MSE by 17.9%.
- **Loss-driven early stopping**: allocates gradient budget proportionally to batch difficulty.

Checkpoint reuse is a valuable supplementary mechanism for specific datasets with recurring patterns, but not the primary source of overall advantage.

6. S6: Dataset details

Synthetic dataset descriptions:

- `synth_stable`: Stationary baseline (no regime changes).
- `synth_trend_break`: Single abrupt trend reversal.
- `synth_slow_drift`: Gradual parameter evolution.
- `synth_fast_switch`: Rapid alternation between 2 regimes.
- `synth_recurring`: 3 regimes that cycle periodically.
- `synth_volatility`: Smooth volatility regime transitions.
- `synth_shock_recovery`: Sudden shock followed by recovery.
- `synth_multi_regime`: 4+ regimes with complex transitions.

Table 5: Dataset summary.

Dataset	Source	Rows	Frequency	Season length
ETTh1, ETTh2	Zhou et al. (2021)	17,420	Hourly	24
ETTm1, ETTm2	Zhou et al. (2021)	69,680	15-min	96
Weather	Wu et al. (2021)	52,696	10-min	144
Exchange	Lai et al. (2018)	7,588	Daily	5
synth_* (8)	Generated	10,000–25,200	Synthetic	50

7. S7: Evaluation protocol justification

Why streaming evaluation.. We evaluate under the streaming protocol because it reflects production deployment: data arrives in discrete batches, the system updates and forecasts before the next batch, and adaptation decisions accumulate. The sliding-window protocol used by Grover and Etemad (2025) and Kim et al. (2025) structurally neutralizes RG-TTA's contributions: (i) checkpoint reuse becomes counterproductive because the model changes by only one step between windows, (ii) early stopping is irrelevant with a 1-step budget, and (iii) regime memory cannot build because consecutive windows overlap by >99%.

DynaTTA fairness.. We acknowledge the protocol mismatch and take three steps:

1. **Horizon-independent warmup**: warmup = warmup_factor×tta_steps× 3, not forecast horizon.
2. **Same frozen-backbone paradigm**: DynaTTA's dynamic LR is applied to the same output projection layer used by all policies.
3. **Transparent reporting**: DynaTTA's per-dataset results are reported alongside ours.

We use the published hyperparameters ($\eta = 0.1$, $\kappa = 1.0$) to ensure faithful comparison, even though increasing η might help under the batch protocol.

8. S8: Retrain comparison details

Per-architecture breakdown.. On GRU-Small, RG-EWC achieves median 34% MSE reduction vs retrain (91% win rate); on iTransformer, median 40% (84%); on PatchTST, retrain wins 61% (+16% median). PatchTST's larger output head benefits from extended gradient budget.

Per-dataset highlights.. RG policies dominate ETTh2 (−42%, 100% wins), ETTm1 (−46%, 100%), Exchange (−57%, 100%). Retrain's advantage is concentrated on synth_volatility (+40% for RG, retrain wins 83%) and PatchTST on synthetic data.

Speed–accuracy.. RG-TTA runs 15–30× faster than retrain. Even RG-EWC is $\sim$16× faster while achieving lower MSE on 70% of configurations.

9. S9: Full hyperparameter configuration

Table 6: Full hyperparameter configuration across all policies.

Component	Parameter	Value
Data Pipeline		
	Sequence length L	96
	Forecast horizons H	96, 192, 336, 720
	Batch size (streaming)	750
	Initial training size	720
	Max batches	10
	Normalization	MinMax $[-1, 1]$
Regime Matching (RG-TTA, RG-EWC, RG-DynaTTA)		
	Feature vector	5-D: $(\mu, \sigma, \gamma_1, \kappa{-}3, r_1)$
	Similarity ensemble	KS (0.3) + Wasserstein (0.3) + Feature (0.2) + VarRatio (0.2)
	Checkpoint loading threshold	sim ≥ 0.75
	Loss gate	$\ell_{\text{ckpt}} < 0.70 \cdot \ell_{\text{curr}}$ ($\geq$30% improvement)
	Memory capacity	5 entries (FIFO eviction)
RG-TTA Adaptation		
	α_{base}	3×10^{-4}
	γ (LR similarity scale)	0.67
	$K_{\max}$ (max steps)	25
	$K_{\min}$ (min steps before early stop)	5
	Patience	3 consecutive steps
	ϵ (min relative improvement)	0.005
	Backbone	Frozen (only output_projection trainable)
EWC (standalone and RG-EWC)		
	λ	400.0
	Fisher samples	200
	Fisher clamp	$[0, 10^4]$
	Online Fisher decay	0.5
DynaTTA (standalone and RG-DynaTTA)		
	$\alpha_{\min}/\alpha_{\max}$	10^{-4} / 10^{-3}
	κ (sigmoid steepness)	1.0
	η (EMA smoothing)	0.1 (published value)
	RTAB / RDB buffer sizes	360 / 100
Baselines		
	TTA: steps / LR	20 / 3×10^{-4}
	EWC: steps / LR / λ	15 / 3×10^{-4} / 400
	DynaTTA: steps	20

10. S10: Reproducibility

Streaming benchmark..

```
PYTHONPATH=.:src:benchmarks \
  python benchmarks/run_unified_benchmark.py \
  --policies tta ewc dynatta \
    rgtta rgtta_ewc rgtta_dynatta \
  --seeds 3 --horizons 96 192 336 720 \
  --models gru_small itransformer \
    patchtst dlinear
```

Dependencies: Python $\geq$ 3.9, PyTorch $\geq$ 2.0, scipy, statsmodels, pandas, scikit-learn. Full list in `pyproject.toml`. All experiments run on CPU; the full 672-experiment benchmark completes in $\sim$48 hours on a 4-core VM.

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